

KOLDING VAMDRUP AP, DenmarkWMO: **061080**Lat: **55.438N**Lon: **9.334E**Elev: **143**StdP: **14.62**Time Zone: **1.00 (EUC)**Period: **00-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.1	21.6	12.2	10.3	21.1	16.7	12.8	23.9	45.8	41.9	38.1	43.5	5.4	320	0.668

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	16.2	79.8	65.3	76.6	64.0	73.3	62.2	67.2	75.9	65.4	73.3	63.8	70.7	10.1	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.2	90.2	70.7	62.5	85.2	68.6	61.0	80.6	66.7	31.6	76.0	30.3	73.4	29.2	70.9	73.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 34.0	27.6	23.4	9.0	84.1	8.3	3.3	3.1	86.5	-1.8	88.4	-6.4	90.2	-12.4	92.6		
(5)			8.0	70.6	7.9	1.7	2.3	71.8	-2.3	72.8	-6.7	73.8	-12.5	75.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	47.8	34.2	34.8	38.7	46.2	52.6	57.8	61.7	62.0	56.7	49.0	41.5	37.4
	DBStd	11.50	7.03	6.22	5.85	5.68	5.65	5.08	5.09	4.67	4.63	5.55	5.93	7.49
	HDD50	2184	490	426	349	141	35	3	0	0	4	85	259	392
	HDD65	6349	955	846	814	564	385	224	132	119	252	496	705	857
	CDD50	1378	0	0	0	27	117	237	363	371	205	54	4	0
	CDD65	69	0	0	0	0	2	8	31	25	3	0	0	0
	CDH74	550	0	0	0	1	15	82	272	167	13	0	0	0
	CDH80	71	0	0	0	0	0	10	41	20	0	0	0	0
(14) Wind	WSAvg	10.9	12.2	13.3	12.5	11.3	11.1	10.7	8.9	9.2	9.3	10.3	10.9	11.3
(15) Precipitation	PrecAvg	33.9	3.1	2.4	2.1	1.6	2.0	2.7	2.9	3.7	3.3	3.6	3.0	3.3
	PrecMax	42.6	6.4	5.6	4.3	4.3	3.6	5.6	6.4	7.7	6.9	8.4	5.9	7.0
	PrecMin	23.4	0.3	0.7	0.4	0.2	0.4	0.1	0.4	1.3	0.7	1.5	0.7	0.9
	PrecStd	5.0	1.5	1.1	1.1	0.9	0.9	1.4	1.5	1.6	1.7	1.3	1.3	1.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.3	50.3	57.4	70.6	75.6	81.7	84.6	82.6	75.7	64.2	57.0	52.2
		MCWB	47.5	45.3	47.2	56.7	61.7	65.3	66.5	66.0	63.5	59.0	54.8	50.1
	2%	DB	46.8	47.1	52.2	64.3	71.7	76.2	80.4	78.6	71.4	60.8	54.0	50.2
		MCWB	45.3	44.9	45.7	52.4	58.3	64.1	65.4	65.0	61.3	56.6	52.0	48.9
	5%	DB	45.1	45.2	48.6	60.3	67.9	72.1	77.1	75.2	67.8	58.8	52.0	48.5
		MCWB	43.4	43.3	44.1	50.9	57.0	61.3	64.4	63.1	59.5	55.3	49.9	47.1
	10%	DB	43.6	43.5	46.4	56.8	64.2	68.4	73.1	71.7	64.7	57.0	50.0	46.7
		MCWB	42.0	41.7	43.4	49.2	55.3	59.5	62.7	62.3	58.3	53.9	48.1	45.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.8	46.8	49.4	58.7	64.3	68.7	69.7	68.7	65.4	59.9	55.3	50.6
		MCDB	48.8	48.4	53.5	68.7	72.6	77.0	78.8	77.7	73.0	62.9	56.7	51.5
	2%	WB	45.5	45.1	46.8	54.2	60.9	65.4	67.5	66.3	62.7	57.6	52.7	49.1
		MCDB	46.5	46.5	50.0	61.1	67.8	73.8	77.1	73.9	68.7	60.0	54.0	49.9
	5%	WB	43.7	43.6	45.1	51.6	58.1	62.8	65.6	64.9	61.1	55.8	50.3	47.4
		MCDB	44.8	45.1	47.8	58.2	65.4	70.2	74.2	71.9	65.9	58.0	51.7	48.3
	10%	WB	42.1	41.8	43.6	49.6	55.8	60.5	63.8	63.6	59.7	54.4	48.3	45.5
		MCDB	43.3	43.2	46.1	55.3	63.0	66.6	70.8	69.8	63.8	56.6	49.7	46.5
(35) Mean Daily Temperature Range	5% DB	MDBR	7.9	8.1	11.4	15.1	16.4	16.1	16.5	16.2	14.3	10.8	8.8	7.7
		MCDBR	7.3	9.0	15.1	21.5	23.1	22.7	22.8	21.4	18.6	12.4	8.8	6.9
		MCWBR	7.6	7.4	10.0	12.0	12.8	12.7	10.7	9.9	10.2	9.2	8.1	6.9
	5% WB	MCDBR	7.2	8.5	12.7	19.1	20.1	19.9	20.1	18.0	15.1	10.8	8.4	6.8
		MCWBR	7.6	7.6	9.3	11.5	12.2	12.3	10.6	9.7	9.2	9.0	8.2	7.0
(40) Clear-Sky Solar Irradiance	taub		0.320	0.339	0.364	0.397	0.389	0.384	0.400	0.400	0.378	0.356	0.334	0.307
	taud		2.248	2.285	2.273	2.231	2.309	2.347	2.321	2.332	2.379	2.383	2.308	2.241
	Ebn at Noon		185	230	252	260	271	273	266	257	247	221	180	161
	Edh at Noon		18	25	32	38	38	37	37	35	29	23	17	15
(44) All-Sky Solar Radiation	RadAvg		157	352	770	1286	1676	1768	1651	1322	917	476	194	115
	RadStd		26	49	71	150	165	138	175	95	95	57	30	12

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (50 neighbors)	+0.67	N/A	N/A	N/A	N/A	N/A	N/A	-283	N/A	N/A

Nomenclature: See separate page